

Table 1: Baseline YOLO test results (mean $\pm$ std).

Model	mAP50	mAP50-95	Params (M)	GFLOPs
YOLOv5				
YOLOv5-n	89.61 $\pm$ 1.33	32.27 $\pm$ 0.63	2.7	7.8
YOLOv5-l	91.11 $\pm$ 0.98	33.01 $\pm$ 0.22	53.2	135.6
YOLOv5-x	90.35 $\pm$ 0.55	32.97 $\pm$ 0.45	97.3	247.3
YOLOv8				
YOLOv8-n	90.02 $\pm$ 0.81	32.62 $\pm$ 0.51	3.2	8.9
YOLOv8-l	89.92 $\pm$ 0.77	32.87 $\pm$ 0.22	43.7	165.7
YOLOv8-x	90.65 $\pm$ 1.86	33.23 $\pm$ 0.90	68.2	258.5
YOLOv9				
YOLOv9-t	90.56 $\pm$ 1.72	32.93 $\pm$ 0.51	2.1	8.5
YOLOv9-c	90.11 $\pm$ 0.89	32.98 $\pm$ 0.50	25.6	104.0
YOLOv9-e	90.29 $\pm$ 1.76	32.85 $\pm$ 1.08	58.2	193.0
YOLOv10				
YOLOv10-n	88.00 $\pm$ 0.89	33.33 $\pm$ 0.49	2.8	8.7
YOLOv10-l	87.62 $\pm$ 1.01	33.38 $\pm$ 0.96	25.9	127.9
YOLOv10-x	87.54 $\pm$ 0.34	33.66 $\pm$ 0.77	31.8	171.8
YOLO11				
YOLO11-n	90.09 $\pm$ 1.49	32.60 $\pm$ 0.98	2.6	6.6
YOLO11-l	89.59 $\pm$ 0.60	33.40 $\pm$ 0.11	25.4	87.6
YOLO11-x	89.99 $\pm$ 0.54	32.68 $\pm$ 0.45	57.0	196.0